

**4** Sodium and potassium ions are involved in the transmission of nerve impulses along an axon.

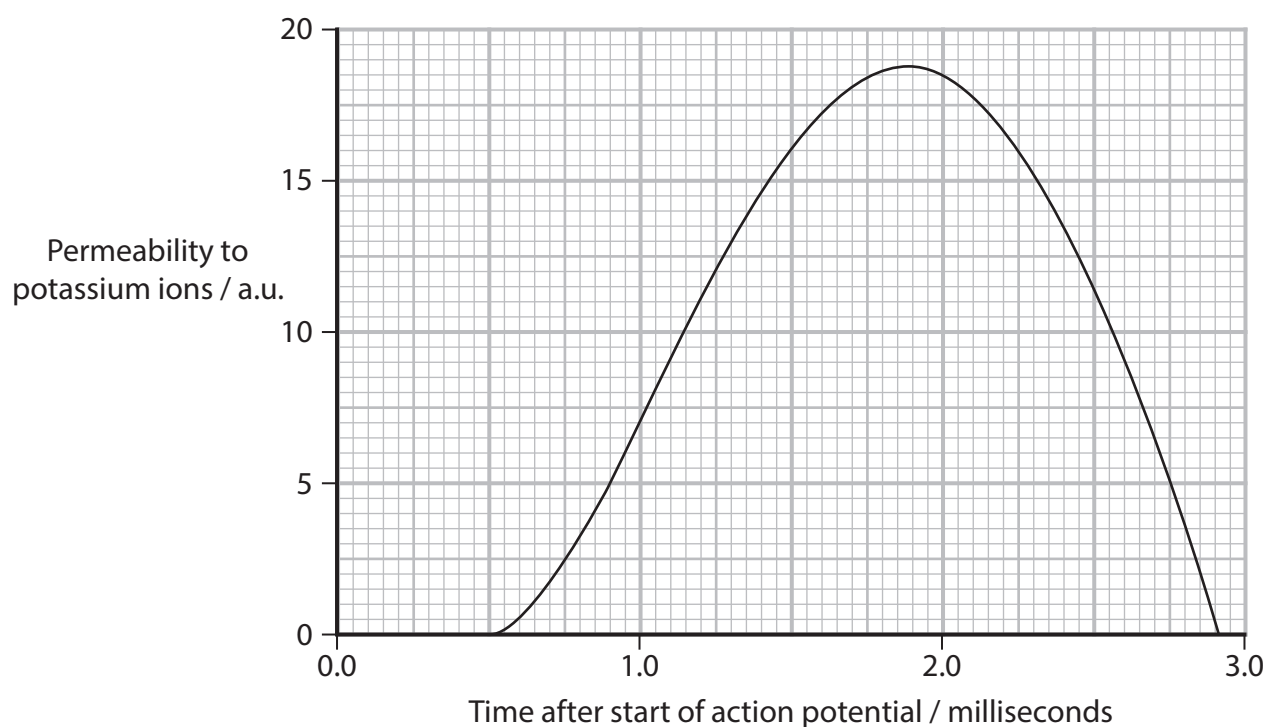
(a) Describe how an action potential is produced in an axon.

(4)

[illegible]

(b) Axon membranes are permeable to potassium ions.

The graph shows the permeability of an axon membrane to potassium ions after the start of an action potential.



Calculate the **maximum** rate of **increase** in permeability to potassium ions.

Give your answer to **two** significant figures with appropriate units.

(2)

Answer .....

- (c) The resting potential in a neurone is the result of the difference in concentration of ions on the outside and inside of the axon when no impulse is being transmitted.

The table shows the concentration of ions outside and inside an axon.

(1)

| Ion                        | Outside concentration<br>/ $\text{mol dm}^{-3}$ | Inside concentration<br>/ $\text{mol dm}^{-3}$ |
|----------------------------|-------------------------------------------------|------------------------------------------------|
| Sodium ( $\text{Na}^+$ )   | 145                                             | 12                                             |
| Potassium ( $\text{K}^+$ ) | 4                                               | 155                                            |
| Chloride ( $\text{Cl}^-$ ) | 120                                             | 4                                              |

Which is the ratio of potassium ion concentration outside and inside an axon?

- ☐ **A** 0.026 : 1
- ☐ **B** 0.3 : 1
- ☐ **C** 0.083 : 1
- ☐ **D** 30 : 1

- (d) (i) Explain the effect of an increase in temperature on the speed of a nerve impulse along an axon.

(2)

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- (ii) The effect of axon diameter and myelination on the speed of an impulse in four neurones was investigated.

The table shows the results of this investigation.

| Neurone | Axon diameter<br>/ $\mu\text{m}$ | Myelination | Speed of impulse<br>/ $\text{ms}^{-1}$ |
|---------|----------------------------------|-------------|----------------------------------------|
| A       | 2.5                              | yes         | 7.0                                    |
| B       | 2.5                              | no          | 3.8                                    |
| C       | 1.5                              | yes         | 4.0                                    |
| D       | 1.5                              | no          | 2.5                                    |

Describe **two** conclusions that can be made from this data.

(2)

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(Total for Question 4 = 11 marks)